Sarah Connor

AI Security Engineer · Adversarial ML & Red Teaming
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SUMMARY

AI security engineer with six years spanning applied machine learning, security engineering, and detection engineering. Builds defenses against adversarial and prompt-injection attacks on production ML systems, and leads red-team exercises against internal LLM deployments.

EXPERIENCE

Senior AI Security Engineer
2023 – Present
Cyberdyne Systems — Austin, USA

Leading red-team engagements against internal LLM-powered products before launch.

- Built an automated prompt-injection fuzzing harness that surfaced 40+ high-severity jailbreaks across 6 production LLM features prior to release.
- Led adversarial robustness reviews for a fraud-detection model serving 2M inference requests/day, reducing evasion success rate from 18% to under 2%.
- Designed and shipped a model-extraction detection pipeline (query-pattern anomaly scoring) now running across all customer-facing inference endpoints.

Python · PyTorch · Burp Suite · Kubernetes · OPA/Gatekeeper

Security Engineer, Applied ML
2021 – 2023
Skynet Analytics — Remote

Owned the security review process for the ML platform team's model deployment pipeline.

- Authored the org's first threat model for LLM-integrated services, adopted as the standard review template company-wide.
- Cut mean time to patch critical model-serving CVEs from 21 days to 4 by embedding security gating directly into the MLOps CI/CD pipeline.
- Ran quarterly red-team exercises against internal chatbots, discovering data-exfiltration paths later closed via output filtering and scoped tool permissions.

Terraform · AWS · MLflow · Semgrep · Snyk

Machine Learning Engineer
2019 – 2021
Tyrell ML — Ljubljana, Slovenia

Built and shipped machine learning models powering fraud detection and demand forecasting for a fast-growing fintech product.

- Built and shipped fraud-detection models into production, reducing false-positive rates by 25% across the company's core payments pipeline.
- Designed the team's first automated retraining pipeline, cutting model refresh time from weeks to days.

TensorFlow · scikit-learn · Docker

SKILLS

Security	Threat modeling, Burp Suite, Nmap, Metasploit
ML/AI	PyTorch, vLLM, adversarial robustness, prompt-injection defense
Infra	Kubernetes, Terraform, AWS, CI/CD pipeline hardening
Practice	Incident response, vulnerability research, red-team exercise design

EDUCATION

M.S. Computer Security 2023
Stanford University Stanford, USA

B.S. Computer Science 2019
University of Ljubljana Ljubljana, Slovenia

PROJECTS

Prompt Injection Firewall 2023
github.com/sconnor/promptfirewall

CERTIFICATIONS, AWARDS & PUBLICATIONS

OSCP – Offensive Security Certified Professional 2019
Offensive Security

AI Village CTF – 1st Place 2024
DEF CON 24, Las Vegas, USA

Detecting Prompt Injection at Scale 2024
arxiv.org/abs/2404.16244